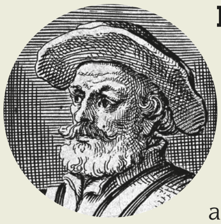
Nobel, Alfred (1833–1896)
Swedish chemist who invented dynamite and smokeless gunpowder. In his will, he donated the majority of his vast fortune to creating the Nobel Prize, an award for achievements in physics, chemistry, physiology or medicine, literature, and peace.

Papin, Denis (1647–c 1712)
French-born British physicist and inventor whose work with steam led to the development of steam engines. Papin was also responsible for inventing the pressure cooker, a steam safety valve, a condensing pump, and a paddle-wheel boat.

Paracelsus (1493–1541)
Swiss–German physician, philosopher, botanist, and astrologer who established the use of chemistry in treating disease. Traveling and practicing medicine across Europe, Paracelsus introduced sulfur, lead, and mercury as remedies for illness.

Pasteur, Louis (1822–1895)
See pp.142–143

Plato (427 BCE–347 BCE)
Greek philosopher and pupil of the philosopher Socrates. In 388 BCE, Plato set up a school in Athens known as the Academy. He set out his theories on how to rule a perfect society in his book *The Republic*. Plato believed all substances were composed of air, earth, fire, and water. He also believed in a spherical Earth and the movement of planets.



Ptolemy (c 100–c 170 CE)
Greek-Roman astronomer, mathematician, and geographer. He built a model of the solar system that explained the movement of the planets and suggested that Earth was at the center of the Universe. He also made a map of the world and wrote an encyclopedia called *Almagest*.

Pythagoras (580–500 BCE)
Greek philosopher and mathematician who influenced the work of Plato and Aristotle. Pythagoras taught that nature and the world could be interpreted through numbers. He is best known for his Pythagorean theorem of geometry and his work on right-angled triangles.

Ramsay, William (1852–1916)
Scottish chemist awarded the 1904 Nobel Prize in Chemistry for discovering the gases argon, neon, xenon, and krypton. He also demonstrated that these gases, along with helium and radon, formed a family of new elements called the noble gases.

Richter, Charles (1900–1985)
American physicist who developed the Richter scale, which measures the magnitude of an earthquake at its epicenter. Richter also devised a map showing the most earthquake-prone areas in America.

Röntgen, Wilhelm (1845–1923)
German physicist who received the first Nobel Prize in Physics in 1901 for his discovery of X-rays in 1895. The introduction of X-rays revolutionized both medicine and modern physics. Röntgen is also known for his discoveries in mechanics, heat, and electricity.

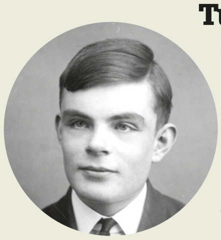
Salk, Jonas Edward (1914–1995)
American physician who discovered the first effective vaccine for polio. Salk began human trials of his polio vaccine in 1952. In 1955, the vaccine was released for widespread use in America.

Shockley, William Bradford (1910–1989)
American physicist who shared the 1956 Nobel Prize in Physics with John Bardeen and Walter Brattain for inventing the transistor, considered one of the greatest breakthroughs in technological history.

Sørensen, Søren Peder Lauritz (1868–1939)
Danish biochemist who introduced the pH scale as a measure of acidity. The scale measures the acidity of a substance either with pH meters or with indicator papers (or solutions) that change color in acid or alkaline substances.

Tesla, Nikola (1856–1943)
See p.155

Thomson, Joseph John (1856–1940)
British physicist who discovered the electron and developed the mathematical theory of electricity and magnesium. He received the 1906 Nobel Prize in Physics for his study of the conduction of electricity through gases.



Turing, Alan (1912–1954)
British mathematician, widely regarded as the father of computer science. During World War II he developed a code-breaking machine known as the Bombe, a prototype for electronic computers, which enabled the British to crack the Nazi code.

Watt, James (1736–1819)
British engineer, whose improvements in steam engine technology contributed to the Industrial Revolution. While repairing a model steam engine, he realized that the engine could be improved by having two cylinders, making them much more powerful.

White, Gilbert (1720–1793)
British naturalist, clergyman, and author who became interested in the natural history around his home in Hampshire, England. In 1789, he published *The Natural History and Antiquities of Selborne*, a collection of correspondence with other naturalists that is still widely read today.